



ASHESI UNIVERSITY

UNDERGRADUATE THESIS

**ACCESS TO URBAN TRANSPORTATION, DELIVERY, AND MARKET
SERVICES IN NIGER (2022–2026): EXPLORING THE NEED FOR AN
INTEGRATED MULTI-SERVICE DIGITAL PLATFORM**

A research Prospectus

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A Research Prospectus Submitted in Partial Fulfillment of the Requirements for SOAN325: Research
Methods

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Abstract

Niger has experienced rapid mobile connectivity growth, rising from approximately 14.6 million connections in 2023 to an estimated 16.5 million by 2025 (Kemp, 2025). Despite this expansion, essential urban services transportation, delivery, and local market transactions remain largely informal and fragmented in Niamey, Niger's capital. This study examines how urban residents accessed these services between 2022 and 2026 and explores the potential of an integrated multi-service digital platform to improve efficiency and accessibility. Grounded in positivism and guided by the Technology Acceptance Model (Davis, 1989) and the Unified Theory of Acceptance and Use of Technology (Venkatesh et al., 2003), the study adopts a quantitative descriptive-explanatory design. Structured questionnaires will be administered to 80–120 urban residents in Niamey. Data will be analyzed using descriptive statistics and regression analysis to identify key predictors of platform adoption intention. Findings are expected to provide a conceptual foundation for designing a context-appropriate digital solution for Niger's informal urban service environment.

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Chapter 1: Introduction

1.1 Background of the Study

The digitalization of urban service delivery has gained substantial momentum across Sub-Saharan Africa as mobile phone penetration and internet connectivity have expanded rapidly over the past decade. Mobile technologies have been shown to reduce information asymmetries, lower transaction costs, and improve coordination across markets (Aker & Mbiti, 2010). In several African cities, ride-hailing platforms such as Uber and Bolt have transformed transportation by enabling users to request, track, and pay for services digitally, improving efficiency and pricing transparency compared to informal alternatives (Acheampong, Osae, & Mekonnen, 2020). Similar trends have emerged in delivery services and mobile commerce, driven by growing smartphone access and mobile money adoption (Demirgüç-Kunt et al., 2018).

Niger has followed this broader connectivity trajectory. According to GSMA Intelligence data, Niger recorded approximately 14.6 million mobile connections in 2023, representing 54.7% of the population (Kemp, 2023). By 2025, this had grown to an estimated 16.5 million connections or 60% of the population (Kemp, 2025), while mobile cellular subscription density reached 65.7 subscriptions per 100 inhabitants in 2023 (International Telecommunication Union, 2024). Yet despite this growth, essential urban services in Niamey Niger's capital remain largely informal. Transportation relies on unregistered street taxis and moto-taxis operating without digital coordination. Delivery arrangements depend on personal contacts, and market transactions occur through informal networks lacking transparent pricing

or digital payment integration. This contrast between growing mobile infrastructure and persistent service informality defines the central problem this study investigates.

1.2 Problem Statement

Between 2022 and 2026, Niger's expansion of mobile connectivity created the technical conditions for platform-based digital services. However, access to urban transportation, delivery, and local market transactions has remained predominantly informal, fragmented, and manually coordinated. Urban residents in Niamey continue to rely on systems characterized by inconsistent pricing, limited reliability, and absent digital coordination conditions that constrain economic participation for informal workers, small business owners, and daily commuters.

This situation raises an analytically important question: given documented mobile connectivity growth, why have integrated digital service platforms not emerged in Niger? The problem involves a complex interaction of economic structures, institutional constraints, user trust, and digital readiness that existing research has not adequately examined. Most literature on digital platforms in Sub-Saharan Africa is concentrated in digitally advanced environments Ghana, Kenya, and Nigeria and focuses on single-service platforms such as ride-hailing or financial technology (Acheampong et al., 2020; Ofori, Anyigba, & Adeola, 2022). Research examining multi-service platform adoption in Sahelian contexts such as Niger is extremely limited, constraining both academic understanding and practical innovation in service digitalization. This study therefore examines how urban residents in Niger have accessed these services between 2022 and 2026 and explores how an integrated multi-service digital platform could address existing inefficiencies.

1.3 Aim and Objectives

The aim of this study is to examine patterns of access to urban transportation, delivery, and local market services in Niger between 2022 and 2026, and to explore how an integrated multi-service digital platform could improve service efficiency and accessibility. This aim is pursued through the following objectives:

- a. To examine how urban residents in Niger accessed transportation, delivery, and local market services between 2022 and 2026.
- b. To assess the role of mobile phone adoption and internet connectivity in shaping patterns of urban service access during the study period.
- c. To identify the key challenges of existing informal systems and determine the functional requirements for an effective integrated digital platform.
- d. To propose a conceptual framework for a unified platform integrating transportation, delivery, and market services relevant to Niger's urban context.

1.4 Research Questions

Central question: How can an integrated multi-service digital platform improve access to urban transportation, delivery, and local market services in Niger?

This central question is supported by the following sub-questions:

- e. How have urban residents in Niger accessed transportation, delivery, and local market services between 2022 and 2026?
- f. What challenges do users face in existing informal service systems, and what features are necessary for a viable integrated digital platform?
- g. How has mobile technology adoption influenced patterns of urban service access?
- h. Under what conditions could an integrated digital platform improve efficiency, coordination, transparency, and accessibility of urban services in Niger?

1.5 Significance of the Study

Academically, this study addresses a geographic and thematic gap in the digital platform literature by examining a Sahelian context where mobile growth has outpaced service digitalization an understudied pattern with theoretical implications for understanding platform emergence conditions. Practically, it provides evidence-based guidance for technology developers and entrepreneurs designing digital solutions for Niger. At the policy level, findings may assist government agencies and international development organizations in identifying the infrastructural and behavioral conditions that support platform adoption. The study also contributes to research linking mobile technology to urban economic development in West Africa, offering a transferable conceptual model for comparable Sahelian contexts.

1.6 Conclusion

This chapter has established the contextual foundation of the study by documenting Niger's mobile connectivity growth alongside persistent urban service informality, and has presented the research problem, aims, objectives, questions, and significance. Chapter 2 reviews the theoretical and empirical literature on digital platform adoption and urban service delivery to provide the analytical foundation for the study's design.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews theoretical and empirical literature related to digital platform adoption, mobile technology, and urban service delivery in Sub-Saharan Africa. The aim is to establish the conceptual and analytical foundation for investigating how an integrated multi-service digital platform could improve urban service access in Niger. The chapter defines key concepts, examines the principal theoretical frameworks, synthesizes relevant empirical evidence, presents a conceptual framework, and identifies the research gaps motivating this study.

2.2 Conceptual Definitions

2.2.1 Digital Platforms

Digital platforms are technological systems that facilitate interactions between multiple user groups through shared digital infrastructure. Parker, Van Alstyne, and Choudary (2016) define platforms as value-creating systems enabling exchanges between producers and consumers rather than producing goods directly. A defining feature is the network effect: platform value increases as participant numbers grow, creating self-reinforcing dynamics that make successful platforms highly scalable. In urban service contexts, platforms reduce friction and information asymmetries inherent in informal service arrangements, enabling real-time coordination between service providers and users.

2.2.2 Ride-Hailing and Multi-Service Platforms

Ride-hailing platforms connect passengers and drivers through GPS-enabled mobile applications, enabling digital payment and service tracking (Acheampong et al., 2020). Research documents improvements in transportation accessibility and pricing transparency in African cities where these platforms operate, alongside income creation for drivers integrated into digital service ecosystems (GIGA, 2021). Multi-service platforms extend this logic by integrating transportation, delivery, and commerce within a single interface. The economic rationale rests on economies of scope: adding service types broadens the user base and increases platform value at declining marginal cost (Parker et al., 2016). Platforms such as Grab in Southeast Asia demonstrate successful multi-service integration in environments initially characterized by informal transportation and commerce, establishing a relevant precedent for comparable African contexts.

2.2.3 Service Informality

Service informality refers to the organization of service provision outside formal regulatory, contractual, and digital frameworks. Informal services operate through personal networks, unwritten agreements, and unregulated pricing, limiting efficiency, accountability, and scalability. In Niger's urban context, service informality characterizes transportation, delivery, and market transactions in ways that define the baseline against which the potential improvements of a digital platform must be assessed. Understanding informality is therefore central to identifying the problem this study addresses.

2.3 Theoretical Foundations

2.3.1 Technology Acceptance Model (TAM)

The Technology Acceptance Model, developed by Davis (1989), is one of the most widely applied frameworks for explaining technology adoption. TAM proposes that adoption intention is primarily determined by two perceptions: perceived usefulness the degree to which users believe technology will improve their performance and perceived ease of use the degree to which they believe it will require minimal effort. Davis (1989) demonstrated that both factors directly influence behavioral intention to use technology, with perceived ease of use also influencing perceived usefulness. The model has accumulated substantial empirical support across diverse technological and cultural contexts (Venkatesh et al., 2003).

In this study, TAM provides the framework for evaluating whether urban residents in Niger are likely to adopt a multi-service digital platform based on their assessments of its utility and usability. If residents perceive a platform as genuinely improving access to transportation, delivery, and market services while remaining straightforward to operate, TAM predicts strong adoption intention. However, TAM's parsimony is a recognized limitation: the model does not explicitly account for external constraints such as infrastructure availability or socioeconomic barriers factors particularly salient in developing country settings. This limitation motivates the complementary use of UTAUT.

2.3.2 Unified Theory of Acceptance and Use of Technology (UTAUT)

Venkatesh, Morris, Davis, and Davis (2003) developed UTAUT as a synthesis of eight competing adoption models to provide a more comprehensive explanation of technology acceptance. UTAUT identifies four core determinants: performance expectancy (the degree to which users believe technology will help achieve performance gains), effort expectancy (perceived ease of use), social influence (the belief that important others think one should use

the technology), and facilitating conditions (the degree to which users believe that organizational and technical infrastructure exists to support use).

The inclusion of facilitating conditions makes UTAUT particularly appropriate for this study. In Niger, infrastructure constraints variable network coverage, limited smartphone penetration in lower-income populations, and restricted mobile financial services access are likely to significantly shape the relationship between adoption intention and actual platform use. Social influence is equally relevant, as community norms in informal economic networks may either accelerate or inhibit the transition to digital platforms. Together, TAM and UTAUT provide complementary analytical tools: TAM focuses on individual perceptual factors, while UTAUT extends the analysis to social and infrastructural dimensions, creating a more comprehensive lens for the complexity of platform adoption in Niger's informal service environment.

2.4 Empirical Review

2.4.1 Digital Platform Adoption in Sub-Saharan Africa

Research on digital platform adoption in Sub-Saharan Africa has grown considerably, though concentrated in a limited set of countries. Acheampong et al. (2020) investigated ride-hailing adoption in Ghana using survey data and regression analysis, finding that convenience, perceived safety, and socio-demographic characteristics are the primary determinants. Younger, more educated, and higher-income residents were most likely to adopt these platforms, suggesting adoption is stratified by socioeconomic status even in markets with adequate mobile infrastructure a finding directly relevant to Niger, where a larger share of urban residents belongs to lower-income categories. Ofori, Anyigba, and Adeola (2022) extended this

analysis in Nigeria, finding that perceived value the assessment of service benefits relative to costs is the dominant predictor of sustained platform engagement. Users who experience service failures or perceive costs as disproportionate are likely to revert to informal alternatives, underscoring the importance of service quality and pricing alignment in cost-sensitive markets.

The GIGA (2021) report documents that ride-hailing platforms in African cities have created measurable income opportunities for informal workers by integrating them into digital service ecosystems, indicating supply-side as well as demand-side benefits from platform adoption. This finding is particularly relevant for Niger's context, where a substantial proportion of transportation and delivery workers operate informally and could benefit from structured digital integration.

2.4.2 Infrastructure, Trust, and Mobile Technology

A consistent finding across the empirical literature is that infrastructure availability and user trust are critical determinants of digital platform adoption. Hornuf, Safari, and Voshaar (2024) examined mobile financial technology adoption across Sub-Saharan Africa and found that trust in digital systems, network coverage quality, and service cost affordability are the strongest predictors of fintech adoption beyond individual perceptual factors. In environments where infrastructure is unreliable, users exhibit elevated sensitivity to service failures, and trust deficits prevent adoption even when mobile access is technically available. Mothobi and Kebotsamang (2024) further demonstrated that network coverage improvements are associated with statistically significant increases in digital technology adoption, confirming that infrastructure enables but does not guarantee platform use, complementary factors including user confidence and perceived benefit must also be present.

Aker and Mbiti (2010) established that mobile phone adoption reduces information asymmetries in markets, enabling faster price discovery and reducing coordination costs across multiple African countries. Mobile-enabled coordination can substitute for formal market institutions in environments where such institutions are weak, a finding particularly pertinent to Niger's informal urban economy. Beyene, Bedemo, and Gebremeskel (2024) extend this analysis at the macro level, finding through panel data that urbanization is a particularly strong predictor of digital technology development, positioning Niamey's urban population as a viable initial target for platform adoption efforts despite Niger's national-level income constraints.

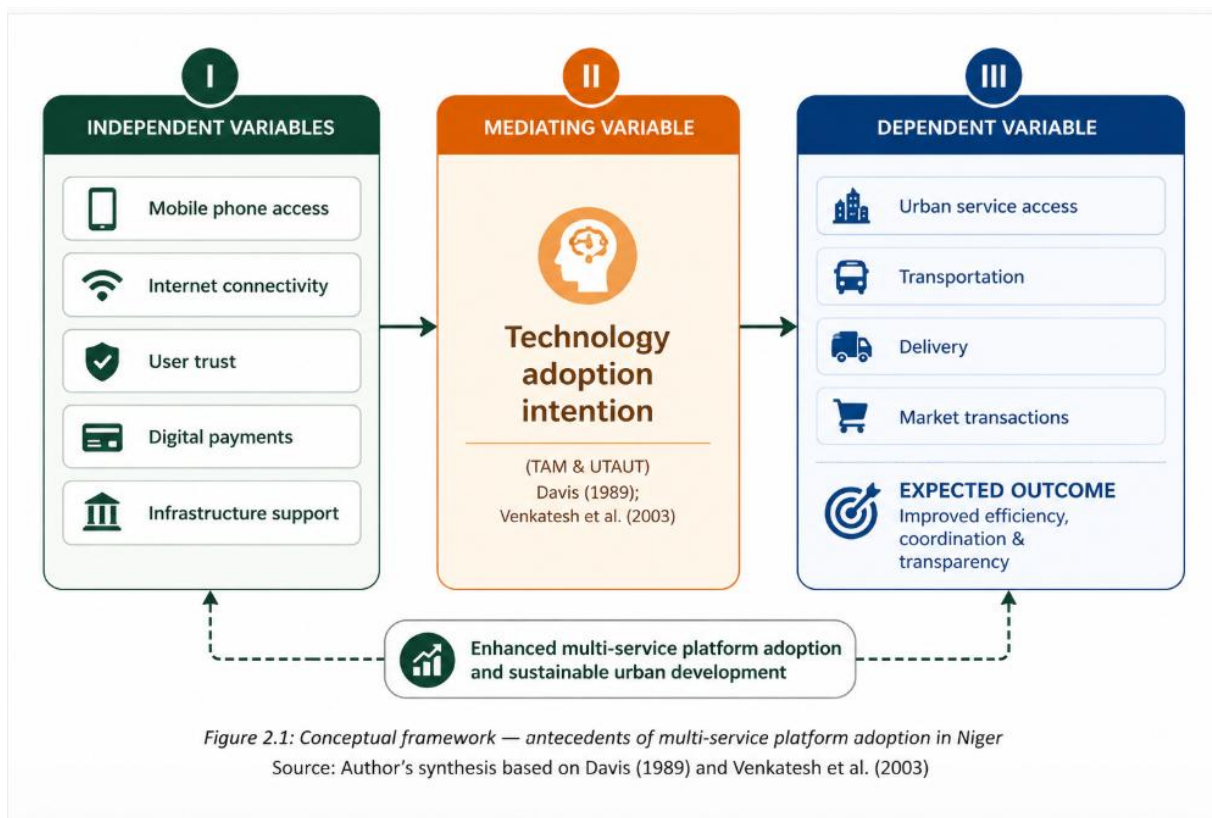
2.4.3 Synthesis and Implications for This Study

The empirical literature reveals that digital platform adoption in Sub-Saharan Africa is shaped by an interaction of individual perceptions, social influences, and structural conditions. Mobile access and infrastructure enable the technical possibility of adoption (Mothobi & Kebotsamang, 2024), while perceived value and trust determine whether this translates into sustained engagement (Ofori et al., 2022; Hornuf et al., 2024). Platform design must therefore address both enabling conditions—reliable access and affordable pricing—and motivational conditions—demonstrating clear utility and building user trust. Critically, most existing studies examine single-service platforms in digitally advanced environments. The specific dynamics of multi-service platform adoption in high-informality, infrastructure-constrained contexts such as Niger remain empirically underexplored, constituting the principal gap this study addresses.

2.5 Conceptual Framework

Based on the theoretical and empirical literature, this study proposes a conceptual framework positioning five independent variables as primary antecedents of technology

adoption intention: mobile phone access, internet connectivity, user trust in digital systems, availability of digital payment systems, and quality of supporting infrastructure. These correspond to TAM's perceived usefulness and facilitating conditions, and UTAUT's performance expectancy and facilitating conditions constructs. Technology adoption intention serves as the key mediating variable linking antecedent conditions to actual platform use. The dependent variable is access to urban services across three domains transportation, delivery, and market transactions with successful platform adoption expected to produce measurable improvements in service efficiency, pricing transparency, and coordination. The framework is illustrated in Figure 2.1.



2.6 Research Gaps and Conclusion

The review identifies three principal gaps motivating this study. First, there is a geographic gap: empirical research on digital platform adoption in Sub-Saharan Africa is concentrated in countries with comparatively advanced digital ecosystems — Ghana, Kenya, Nigeria, and South Africa while Sahelian contexts such as Niger remain substantially underexamined. Second, there is a thematic gap: existing studies examine single-service platforms in isolation, while multi-service platform adoption which requires users to trust and engage with a more complex integrated system has received limited empirical attention. Third, there is a contextual gap: TAM and UTAUT have been predominantly applied in environments where digital alternatives already exist; their predictive power in high-informality settings where formal digital alternatives are largely absent has not been rigorously evaluated.

Together, these gaps establish the need for context-specific empirical investigation of how integrated digital platforms could improve urban service access in Niger. The conceptual framework presented in this chapter structures the causal relationships between key variables and provides the analytical foundation for the methodological approach developed in Chapter 3.

Chapter 3: Research Methodology

3.1 Introduction

This chapter presents the methodological framework guiding this study. It covers research philosophy and reasoning approach, methodological choice and design, research scope, sampling, variable operationalization, instrumentation, data collection and analysis, limitations and validity, and ethical considerations. Each decision is justified with reference to the study's research questions and the theoretical framework from Chapter 2. This study aligns with the Undergraduate Thesis capstone option.

3.2 Research Philosophy and Reasoning Approach

This study is grounded in a positivist research philosophy. Positivism holds that social reality can be investigated through systematic empirical methods based on measurable, observable phenomena (Creswell, 2014). Positivism is appropriate because the central questions concern quantifiable behaviors and attitudes: how urban residents access services and what conditions shape their platform adoption willingness. These phenomena are amenable to statistical analysis, consistent with the measurable constructs of TAM and UTAUT. The study employs deductive reasoning proceeding from the established propositions of TAM and UTAUT to empirical tests of those propositions (Bryman, 2016) ensuring systematic alignment between theory and empirical inquiry.

3.3 Methodological Choice and Research Design

The study adopts a quantitative methodological approach, appropriate when the aim is to measure variables, identify statistical relationships, and evaluate findings against theoretical

expectations (Creswell, 2014). A descriptive and explanatory research design is employed. The descriptive component characterizes how urban residents in Niamey accessed transportation, delivery, and market services between 2022 and 2026. The explanatory component evaluates the causal relationships in the conceptual framework, examining how mobile access, user trust, infrastructure quality, and digital payment availability influence adoption intention. A cross-sectional time horizon is used, collecting data at a single point in time, sufficient to assess current access patterns and adoption readiness within the study's practical constraints.

3.4 Research Scope: Study Area and Target Population

The study focuses on Niamey, Niger's capital and largest urban center, selected because it has the highest concentration of mobile phone users, hosts the most active informal transportation and market service systems, and represents the most viable environment for early digital platform adoption. The target population comprises urban residents who extensively use transportation, delivery, or local market services, specifically: daily commuters using street taxis or moto-taxis; users of informal delivery services; market traders and small business owners; and individuals with mobile phone access and some application experience. These groups represent both current informal service users and likely platform adoption candidates.

3.5 Sampling Method and Sample Size

Given the absence of a comprehensive sampling frame for informal service users in Niamey, this study employs convenience sampling selecting accessible and willing participants. Convenience sampling is widely used in exploration studies of informal economic activity where population registers do not exist (Bryman, 2016). While the primary limitation of non-probability sampling is restricted statistical generalizability, this approach is appropriate for an

exploratory study seeking to identify patterns and generate hypotheses rather than produce population-level estimates. Participants will be recruited at transport hubs, central markets, and commercial areas in Niamey, where informal service activity is naturally concentrated. A target sample size of 80–120 respondents is sufficient to support descriptive statistics and basic regression analysis at a meaningful level of significance (Field, 2018). To mitigate the limitations of convenience sampling, efforts will be made to ensure diversity in the sample by collecting data across multiple locations (transport hubs, markets, and commercial areas) and at different times of the day. This approach increases the likelihood of capturing respondents from varied socioeconomic and occupational backgrounds. Additionally, basic demographic balancing (e.g., gender, occupation categories) will be monitored during data collection to reduce overrepresentation of any single group.

3.6 Operationalization of Variables

The conceptual framework identifies three variable categories. Independent variables include mobile phone access (ownership and internet use frequency), internet connectivity (self-reported quality and reliability), user trust in digital systems (confidence in digital payment security and platform reliability), digital payment availability (current mobile money use), and infrastructure support (perceived network reliability). The mediating variable is technology adoption intention, operationalized through items derived from TAM's behavioral intention construct and UTAUT's performance expectancy, effort expectancy, and social influence constructs. The dependent variable is access to urban services, measured across three dimensions: transportation (frequency, cost, and reliability of current use), delivery (current access methods and perceived adequacy), and market transactions (reliance on informal

networks and interest in digital alternatives). A full operationalization table is provided in Appendix B.

3.7 Instrumentation

Data will be collected through a structured questionnaire organized into five sections. Section A captures demographics (age, gender, education, occupation). Section B documents current urban service access patterns, preferred modes, and perceived challenges with informal systems. Section C assesses mobile and internet access including device ownership, data subscription status, and mobile money use. Section D measures TAM and UTAUT adoption constructs, covering perceived usefulness, ease of use, social influence, and facilitating conditions. Section E captures platform adoption intention, measuring likelihood of using an integrated multi-service platform under various conditions. Sections D and E use five-point Likert scales (1 = Strongly Disagree to 5 = Strongly Agree), consistent with established practice in TAM and UTAUT research (Davis, 1989; Venkatesh et al., 2003). The complete questionnaire is provided in Appendix A.

3.8 Data Collection and Analysis Procedures

Primary data will be collected through face-to-face questionnaire administration, preferred over self-completion given literacy variability in the target population. The researcher will administer questionnaires personally at the identified recruitment locations, ensuring consistent instruction delivery and minimizing non-response errors. Secondary data including academic journals, World Bank and ITU reports, and datasets from DataReportal and GSMA provide contextual background on Niger's digital landscape and support interpretation of primary findings.

Quantitative data will be analyzed using descriptive and inferential statistical methods. Descriptive statistics frequencies, percentages, and mean scores for Likert-scale items will characterize response distributions and profile service access patterns and adoption readiness across the sample. Multiple regression analysis will examine relationships between the independent variables and technology adoption intention, identifying the strongest predictors while controlling demographic characteristics. Statistical significance will be assessed at $p < 0.05$. Descriptive service access data will be further analyzed through frequency distributions and cross-tabulations to identify patterns across demographic sub-groups.

3.9 Limitations, Delimitations, and Validity

The study has three principal limitations. First, convenience sampling restricts statistical generalizability beyond the sample. Second, self-reported data may be subject to social desirability bias, partially mitigated through questionnaire design that emphasizes honesty and anonymity. Third, cross-sectional design captures attitudes at one point in time, limiting insight into how adoption intention evolves into actual behavior. The study is intentionally delimited to Niamey, the most digitally connected and commercially active urban environment in Niger, which is the most relevant site for initial platform adoption inquiry.

Content validity is ensured by systematically aligning all questionnaire items with research objectives and TAM and UTAUT constructs, with faculty review prior to deployment. Internal validity is strengthened through validated scale items from established TAM and UTAUT research. External validity is limited by convenience sampling; however, the study's theoretical grounding in cross-culturally validated frameworks supports cautious transferability of findings to comparable informal urban contexts in the Sahel region.

3.10 Ethical Considerations

This study will be conducted in full compliance with social science research ethical standards. Participation will be entirely voluntary, with no coercion or obligation applied at any stage. Informed consent will be obtained from all participants prior to questionnaire administration, including clear explanation of the study's purpose, participation requirements, data usage, and confidentiality protections. All responses will be anonymous: no personal identifying information will be collected, and data will be stored securely with access restricted to the researcher and supervising faculty member. Data will be used exclusively for the academic purposes of this research prospectus, consistent with Ashesi University's research ethics guidelines.

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Appendices

Appendix A: Research Questionnaire

Research Questionnaire

Study Title: Access to Urban Transportation, Delivery, and Market Services in Niger
(2022–2026)

Instructions: This questionnaire is part of an academic research study at Ashesi University. All responses are completely anonymous and used solely for academic purposes.

Section A: Demographics

A1. Age: ____ years

A2. Gender: Male / Female / Prefer not to say

A3. Education: Primary / Secondary / Tertiary / None

A4. Occupation: _____

Section B: Urban Service Access

B1. Primary travel mode: Personal vehicle / Street taxi / Moto-taxi / Walking / Other:

B2. Frequency of transportation use per week: ____

B3. Satisfaction with current transportation reliability (1–5): ____

B4. How do you access delivery services? _____

B5. Do you use digital apps for any services? Yes / No. If yes:

Section C: Mobile and Internet Access

C1. Smartphone owner? Yes / No

C2. Active mobile data subscription? Yes / No

C3. Mobile internet quality in your area (1 = Very poor, 5 = Excellent): ____

C4. Mobile money services user? Yes / No

Section D: Technology Adoption Perceptions (1 = Strongly Disagree to 5 = Strongly Agree)

Statement	1	2	3	4	5
D1. A digital platform for transportation and delivery would improve my daily life.					
D2. I believe a digital platform would be easy to learn and use.					
D3. People I know would encourage me to use a digital service platform.					
D4. I have sufficient internet access to use a digital platform.					
D5. I would feel confident that my data would be secure on a digital platform.					
D6. I would use a digital platform if pricing were comparable to current services.					

Section E: Platform Adoption Intention

E1. Likelihood of using an integrated platform for transportation, delivery, and market services if available in Niamey (1–5): ____

E2. Most important feature needed: _____

E3. What would prevent you from using such a platform?

Thank you for your participation.

Appendix B: Variable Operationalization Table

Variable	Category	Measurement	Items
Mobile phone access	Independent	Ownership + use frequency	C1, C2
Internet connectivity	Independent	Quality rating (1–5)	C3
User trust	Independent	Digital security confidence	D5
Digital payment availability	Independent	Mobile money use	C4
Infrastructure support	Independent	Network reliability	C3, D4
Adoption intention	Mediating	Likelihood to adopt (1–5)	E1
Transportation access	Dependent	Mode, frequency, satisfaction	B1, B2, B3
Delivery access	Dependent	Current delivery methods	B4
Market access	Dependent	Informal vs. digital use	B5